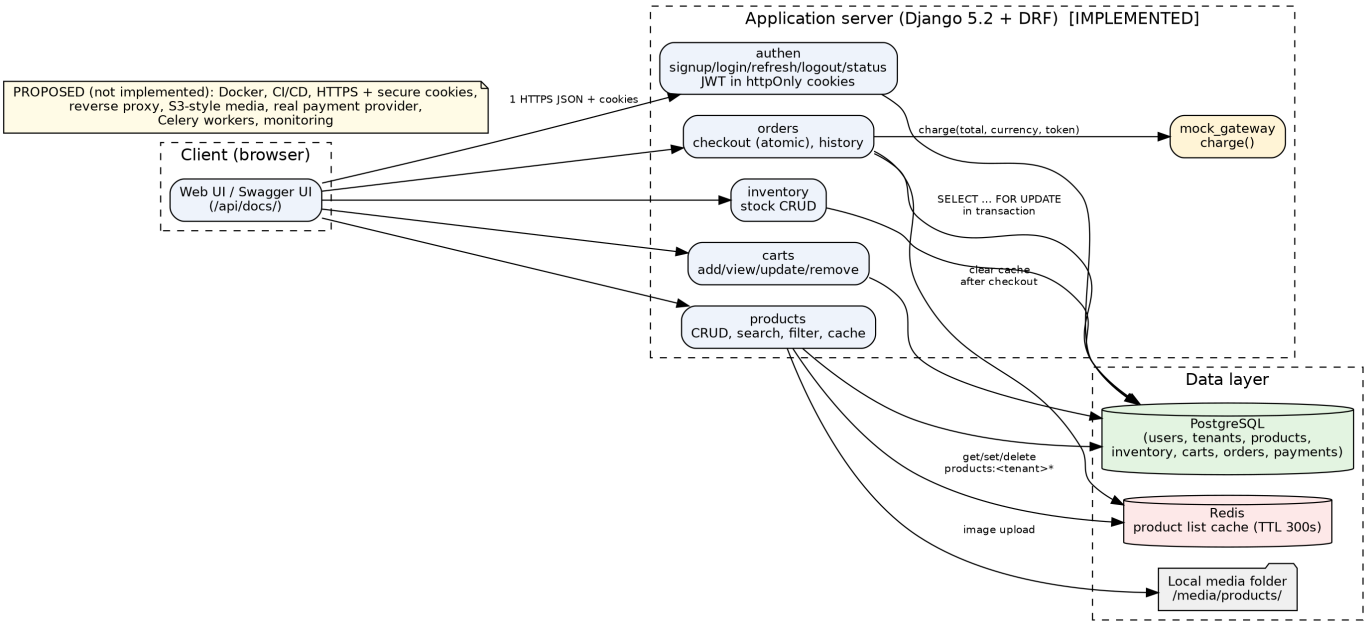
System Architecture - E-commerce Store



Data movement

1. The client sends JSON over HTTP; authentication travels in httpOnly cookies (access 15 min / refresh 7 days). 2. DRF views read/write PostgreSQL through the Django ORM. 3. Product list responses are cached in Redis for 300 s under keys products:<tenant>:... and every write (create/update/delete/checkout) deletes that tenant's keys; Redis failures are swallowed so the site works without it (fail-open). 4. Checkout runs inside one database transaction and calls the mock payment gateway in-process; any failure rolls everything back. 5. Product images are saved to the local media folder.

Boundaries: implemented vs proposed

Component	Status	Notes
Django/DRF API, PostgreSQL, Redis cache, mock gateway, Swagger	Implemented	Runs locally
Docker, CI/CD, HTTPS/secure cookies, reverse proxy, object storage, real payment provider, background workers, monitoring	Proposed only	Not built; see engineering-decisions.md
Multi-tenancy	Partial	Products and carts are tenant-scoped; inventory API is not (D-06)